

Stress loading of the cable shovel boom under in-situ digging conditions

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Received 26 January 2006; accepted 11 February 2006

Available online 30 March 2006

Abstract

Cable shovels are capital-intensive equipment, whose operations are characterized by high stress loading and fatigue failure resulting in high maintenance costs. The operating environment and the stress fields on the boom must continually be monitored to avoid random fatigue failure to achieve reliability, longevity and economic usage. In this paper, the authors develop dynamic models for real-time stress monitoring using a combination of flexible and rigid body approach. A virtual prototype simulator is developed to simulate the cable shovel excavation in oil sands and to examine the motion, stress and local deformation of the boom. The P&H 4100A cable shovel, deployed in the Athabasca oil sands formation, is used to examine the cable shovel boom durability using stress fields simulation. The results show that high stresses occur at regions around the joint between the upper body and the boom, resulting in large deformations. In hard formations, the results show that the stress fields in this region exceed the Von Mises yield strength of steel used in making the shovel boom components. The results also show that the FE nodes of 178, 168, 120, 63, 127 and 84 for 10 MPa modulus of elasticity and 127, 61, 126, 45, 60 and 39 nodes for 20 MPa modulus of elasticity are the highly stressed nodes with high degree of boom deformation and fatigue failure. The study provides a solid foundation for further study of failure life analysis of the cable shovel components. © 2006 Elsevier Ltd. All rights reserved.

Keywords: Cable shovel boom; Kinematics and dynamics; Stress loading and deformation fields; Virtual prototype simulation; Durability analysis

1. Introduction

The cable shovel excavator, shown in Fig. 1a, is widely used as primary production equipment in surface mining operations. Stress loading and fatigue failure in equipment components such as booms cause unplanned downtimes, reduced efficiency and high production costs. The repeated dynamic, instantaneous loading accumulates over the cable shovel service life, which may induce boom damage [1]. The boom may fail if stress loading accumulation reaches a threshold level. Conventional design process uses a laboratory test to simulate the cable shovel front-end assembly for component integrity. This process is very costly and time-consuming for achieving statistically significant basis for evaluation and subsequent decisions. Previous

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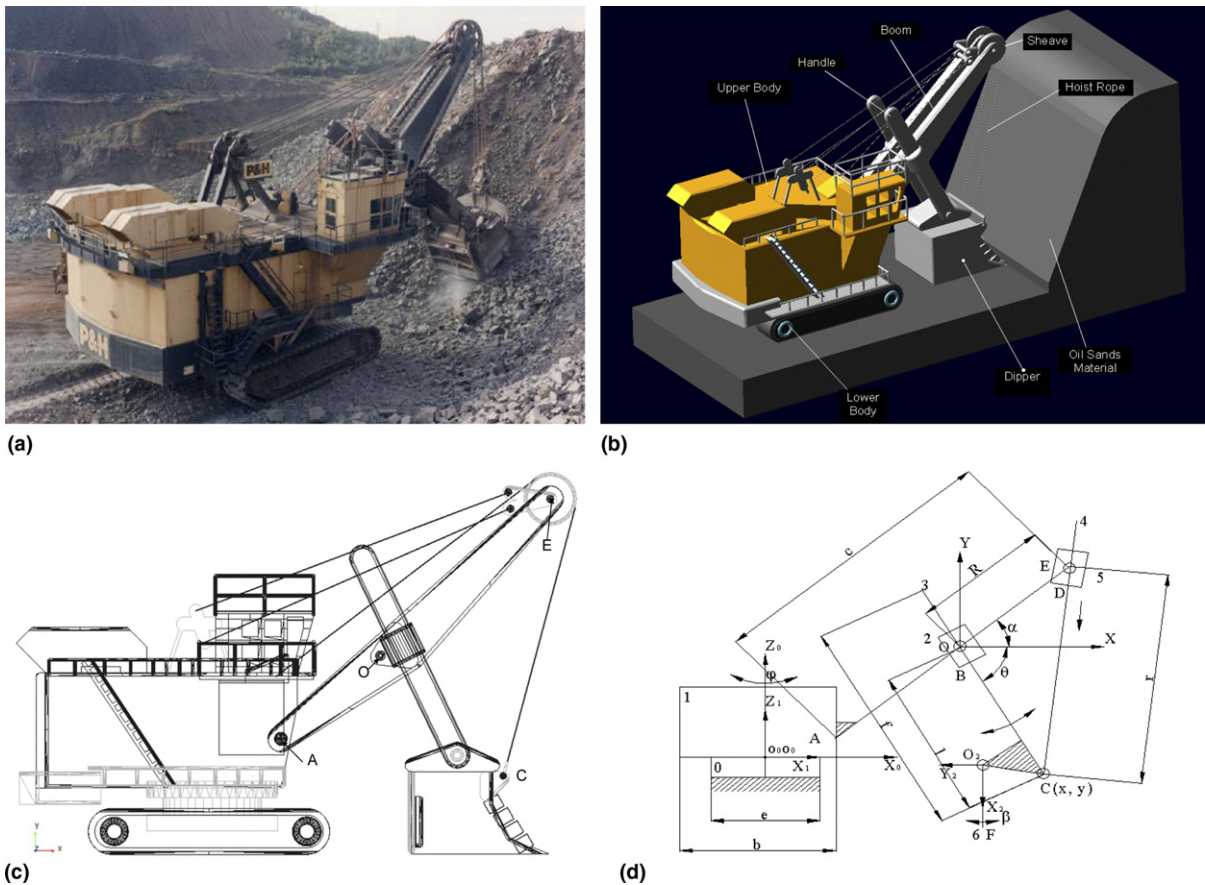


Fig. 1. Cable shovel system: (a) physical prototype, (b) virtual prototype, (c) structures and (d) components assemblage and linkage motions.

research has emphasized the establishment of the mathematical relationships between the driving forces of cable shovel performance parameters and the resistive forces from the working materials during excavation [2,3]. Recently, 2D and 3D virtual shovel prototype simulators have been developed to simulate surface mining processes [5]. Modern design processes in virtual simulation environments enable the testing of machine performance and component fatigue failure prior to final design and field-testing. The major shovel failure modes are deformation and fracture caused by excessive stress in shovel excavation [6]. The finite element method (FEM) can be used for computing stresses and deflections in shovel components. The inputs into such programs must include a complete description of the forces acting on the components via dynamic modeling [7].

Cable shovel is a multi-degree of freedom, non-linear multi-body dynamic system [3]. Thus, multi-body dynamic theory is used to create the kinematics and dynamic models of this mechanical system in this paper [8]. Component stress loading is modeled using flexible multi-body dynamics. Virtual prototype simulation, based on comprehensive machine dynamics, is advanced to simulate the stress loading of the cable shovel front-end assembly during excavation. The virtual simulator is built as a rigid multi-body dynamic model and is used to simulate the spatial kinematics and dynamic performance based on given trajectories. The model does not consider formation variation in the vertical plane and the flexibility of structural components. For accurate fatigue failure prediction, realistic 3D model, with flexible components and various material formations, are required for capturing component stress loading. ADAMS/NASTRAN and ADAMS/FLEX are used to provide solutions to the stress loading equations for the shovel components [9]. These software packages contain computationally efficient numerical simulation routines for executing realistic full-motion behavior of complex mechanical systems and provide quick analysis for multiple design variations toward an optimal design [10]. They are developed for rigid and flexible multi-body systems using Newton–Euler–Lagrange equations and

FEM. An integrated environment of rigid and flexible components, capturing the cable shovel dynamics, is built and the FEM technique is used to find solutions to the stress loading of the shovel components [11].

2. Multi-body models of cable shovel and oil sands

The cable shovel-formation dynamic models are developed using multi-body dynamic theory as a basis for developing the virtual prototype simulators.

2.1. Basic cable shovel structures

Fig. 1b illustrates a multi-body virtual cable shovel prototype model in an oil sands environment. The key components include lower body, upper body and attachment. The attachment consists of the boom, handle, rope and dipper. Fig. 1c displays the cable shovel structure with the main structural components and the boom connections. The boom is mounted on the upper body using center pin at point A. The handle is connected to the boom by a rigid triangular frame joint at point O. The hoist ropes pass over sheaves mounted on the boom at point E. The digging trajectory of the dipper is produced by extending or retracting the handle. The boom, handle, and the hoist cables produce the desired digging action with crowding and hoisting. The crowd action refers to the motion of the dipper along the crowd arm plane. Dipper hoisting is accomplished by means of vertical or near-vertical plane movements.

2.2. Multi-body cable shovel model

The cable shovel is modeled as a seven-bar linkage with four degrees of freedom (DOFs) for studying its kinematics and dynamics as in Fig. 1d [5]. The motion of each shovel component is described in terms of the generalized coordinates of the Euler–Lagrangian formulation for constrained system given by Eq. (A.1).¹ The reactive torques or forces in the cable shovel are given by the summation term in Eq. (A.1) along each of the generalized coordinates. Therefore, component assemblies are represented mathematically in a manner that conforms to the required system dynamic functions.

2.3. Flexible oil sands model

The resistive force on the dipper by the oil sands formation causes boom stress concentration and local boom deformation. The oil sands load-deformation model is created as a continuous system with a finite number of units connected by spherical joints. Fig. 2 shows a 2-DOF model, which has been idealized using a simple mass–spring–dashpot system. The mass is equal to the mass of an oil sand unit and the spring represents its elastic properties. The spring stiffness represents the elastic modulus of the oil sands. The dashpot represents the damping characteristic of the oil sands. The oil sand unit is subjected to a contact force when the dipper moves on it. Based on Newton's second law, the motion-damping-deformation equation for an oil sand unit is formulated as Eq. (A.2) [12]. This deformation can be calculated by mass, elastic modulus and damping characteristic of the oil sand unit (Fig. 2).

2.4. Loading on booms

Fig. 3 shows a free-body diagram of the boom loading mechanism. The xyz coordinate system has been selected with x as horizontal axis and y as vertical axis. The five forces (torques) acting on the boom free-body can be calculated by applying Euler–Lagrange formulation. These forces (torques), which are identified in x , y and z component form, are dead load F_G from the boom weight, force F_E and torque T_E from hoist ropes, force F_H and torque T_H from suspension ropes, force F_O and torque T_O from the handle and force F_A and torque T_A from the upper body. The dynamic force F_E and torque T_E are produced by the dipper acting through the rope.

¹ The detailed formulation of all the mathematical equations is contained in the [Appendix A](#).

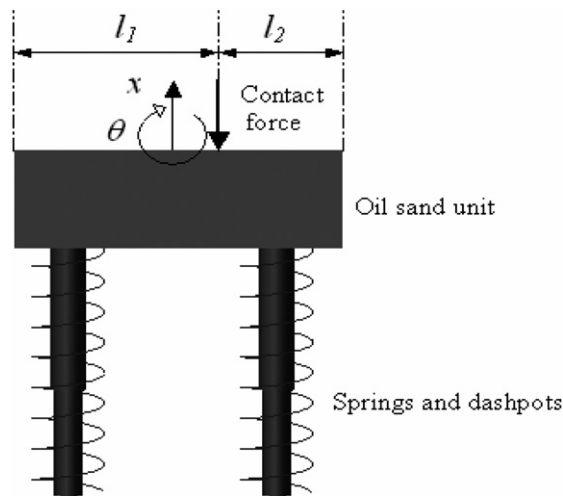


Fig. 2. Load-deformation model.

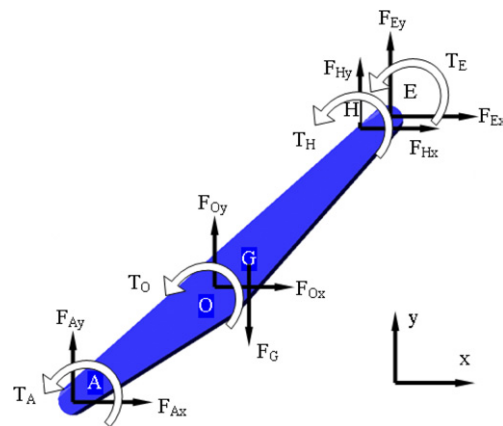


Fig. 3. Free-body diagram of boom.

The dynamic force F_o and torque T_o are produced by the dipper acting through the handle. The ADAMS software is used to simulate the kinematics, static-force and dynamic-force conditioning of the boom.

3. Flexible boom stress recovery

The boom's flexible body model is developed by integrating a finite element flexible body into the rigid multi-body shovel component model. Component mode synthesis is used to reduce the total DOFs of detailed finite element (FE) component stress models, while preserving its local deformations with high level of accuracy. Flexible structural component motion with N DOFs and defined boundaries are described by a combination of P normal and S constraint modes. The structural component of the cable shovel is modeled in dynamic equilibrium as Eq. (A.7). The system DOFs are partitioned into internal and boundary DOFs, as illustrated by Eq. (A.8). At static equilibrium, assuming that interior forces are set to zero, Eq. (A.8) becomes Eq. (A.9). The constraint modes matrix, the normal modes matrix, and the physical coordinates are formulated as Eqs. (A.10), (A.11) and (A.12), respectively. Assuming that the reduction of the full set of mode shapes for the flexible body is correct and captures all the required effects, the stress distribution and component deformation are modeled as Eqs. (A.14) and (A.15), respectively.

Fig. 4 shows an outline of the proposed method and its conceptual data flow, from structural FE analysis to stress history calculation. Modal stress recovery plug-in is used to calculate and display Von Mises stress distribution within ADAMS. Using plug-in recovery, parameter studies and optimization can be performed on

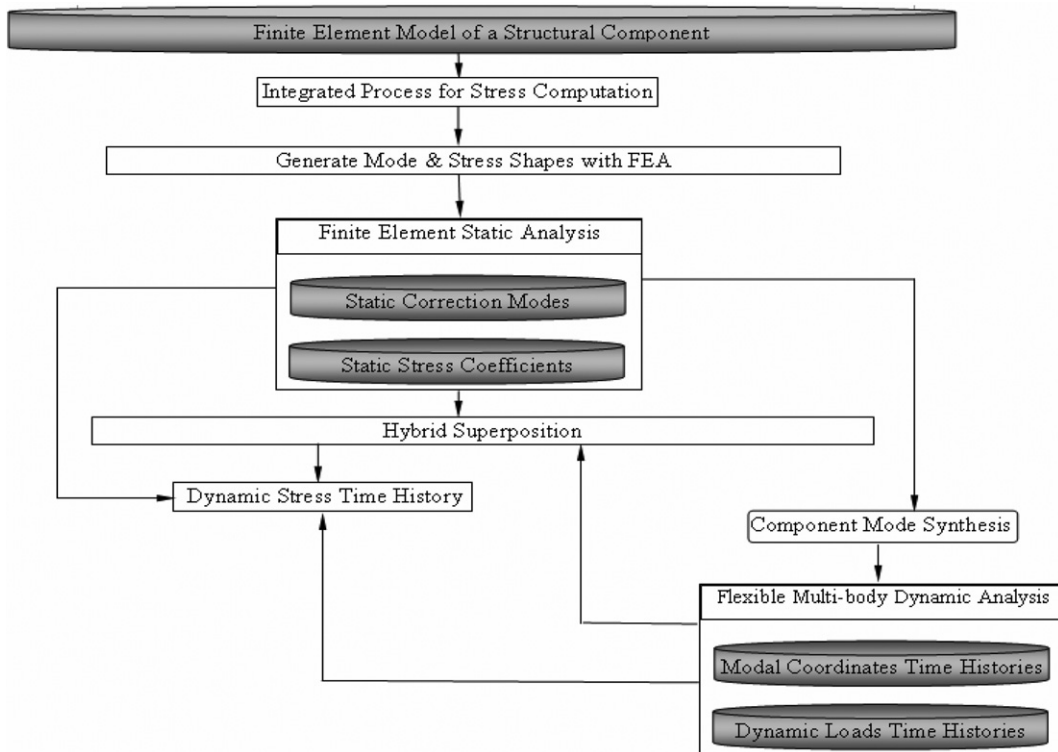


Fig. 4. Conceptual flow of boom stress analysis.

the system. Modal velocity and acceleration stresses are used to calculate dynamic Von Mises stress time histories that consist of pseudo-static stress [13]. The focus of the models is on time-dependent terms, dynamic loads, velocities and accelerations of modal coordinates, and time-independent terms like stress coefficients. Flexible multi-body dynamic analysis is used to calculate dynamic loads and modal coordinates of the structural components. The sub-structuring technique [13] is used for dynamic analysis as the boom is separated from the multi-body cable shovel. Using flexible multi-body dynamic analysis code, the boom is simulated under the specified scenario to obtain dynamic load time histories and modal coordinate time histories. FE static analysis is used to calculate static stress and modal stress coefficients. The FE static analysis is carried out by applying each unit load at an FEM node where corresponding dynamic load is applied. The sub-structuring technique is used to calculate stress coefficients in local FEM model area. Static stress and modal stress influence coefficients are multiplied and summed at each integration time step to yield the dynamic stress time histories at that time step.

4. Stress and deformation modeling in a virtual mine

The P&H 4100A cable shovel is deployed in the Athabasca oil sands formation to excavate in situ materials with little or no pre-fragmentation. The random occurrence of hard formations presents a major problem to shovel excavation [3]. This cable shovel has been used to illustrate a numerical example to demonstrate the practical significance of the study under two assumptions. The assumptions include (i) dipper handle, dipper and body components are rigid with rotation and translation only at joints; and (ii) oil sands are homogeneous.² Table 1 contains the main geometrical dimensions [4] for creating the virtual model of the cable shovel. The mass and inertia properties of the main shovel components [5] are given in Table 2. Table 3 contains the material properties for the boom and oil sands. Two cases of oil sands material are used in this simulation as

² This is a very practical assumption given that the oil sands occur in thick deposits with minimum variations in physical composition.

Table 1
Cable shovel dimensions for a P&H 4100A [4]

General dimensions (see Fig. 5)			
Overall width A (m)	13.16	Overall width of crawler (Std) E (m)	9.86
Overall length B (m)	15.24	Overall length of crawler F (m)	11.58
Overall height over gantry C (m)	12.62	Ground clearance G (m)	1.07
Width of crawler shoe D (m)	2.21	Height-ground to bottom of counterweight H (m)	2.77
Main component data (see Fig. 1d)			
Boom length, c (m)	20.57	Handle length, f (m)	13.19
Boom length, R (m)	13.18	Dipper length ^a (m)	5.8
Boom width (m)	0.77	Hoist(wire rope) diameter (mm)	70
Boom angle α (°)	44.3	Open dipper door angle β (°)	0–90
Swing angle φ (°)	0–90		

^a Length between the joint for handle and dipper and dipper tip.

Table 2
Mass and inertia properties of main shovel components [5]

	Material	Mass (kg)	Inertia moments ^a (kg m ²)
Boom	Steel	164,000	2.252E + 6
Handle	Steel	65,800	7.749E + 5
Dipper	Steel	24,500	6.458E + 4

^a Moment of inertia about the gravitational center.

Table 3
Material properties of the boom and oil sands material

Material property	Scalar value		
	Boom [3]	Oil sands [14] Case 1	Oil sands [14] Case 2
Modulus of elasticity	206.7 GPa	10 MPa	20 MPa
Poisson's ratio	0.30	0.30	0.30
Mass density	7820 kg/m ³	1600 kg/m ³	1600 kg/m ³
Damping constant		~120 kNs/m	~120 kNs/m

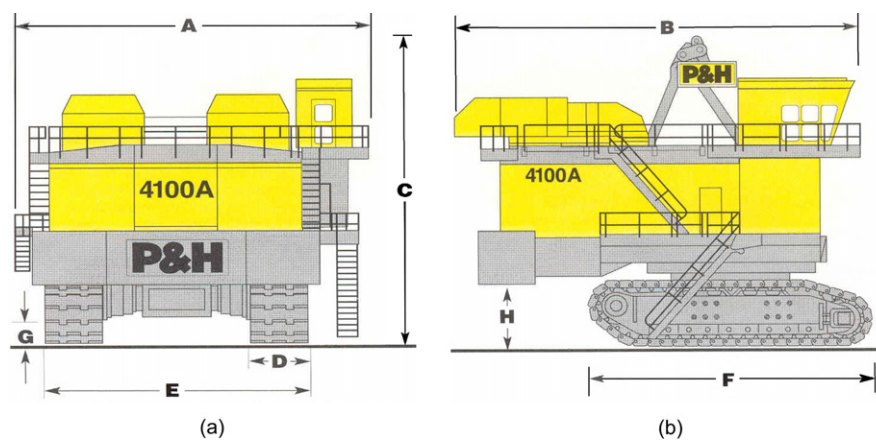


Fig. 5. Cable shovel dimensions for P&H 4100A [4]: (a) back view and (b) side view.

illustrated in Table 3 under Cases 1 and 2. The material properties of the boom and oil sands are obtained at an environmental temperature of 25 °C in Table 3 [3,14]. The actual shovel-formation interactions are recreated into a virtual model and simulated in the ADAMS environment to study the stress loading and deforma-

tion of the shovel boom during digging operations. The boom is modeled using the sub-structuring technique in order to include its structural deformation modes. Flexibility effects are introduced and examined within the rigid boom model. MSC.NASTRAN is used for eigenvalue and static stress analysis with free boundary conditions.

5. Virtual simulation results and analysis

Fig. 6a and b shows the respective periodic joint angular displacements and velocities for shovel motions. The angular displacements are very smooth for each motion, increasing rapidly from minimum to maximum and then staying constant at the maximum. The respective maximum angular displacements (degrees) for the hoist, crowd, swing and dump motions are 30, 60, 90 and 90. The joint angular velocities are also very smooth, increasing rapidly from minimum to maximum and decreasing with almost the same gradients. The respective maximum angular velocities (degrees per second) for the hoist, crowd, swing and dump motions are 12, 20, 28 and 30. The angular displacements and velocities of the hoist motion show the least changes, followed by the crowd mechanism. The swing and dump motions show the highest changes in the time run. These conditions show that the simulator mimics efficient cable shovel operations. The stress loading of the shovel boom is developed using the finite element method (FEM). Fig. 7a and b displays the isometric and shaded views of the boom geometry used for FE analysis. In order to reduce the computational time, a simplified FE mesh is generated using parabolic tetrahedral elements with element size of 20 mm, minimum size of 10 mm, a minimum angle of 10° , a span angle of 20° , a growth rate of 2 and modes number of 6. The actual digging

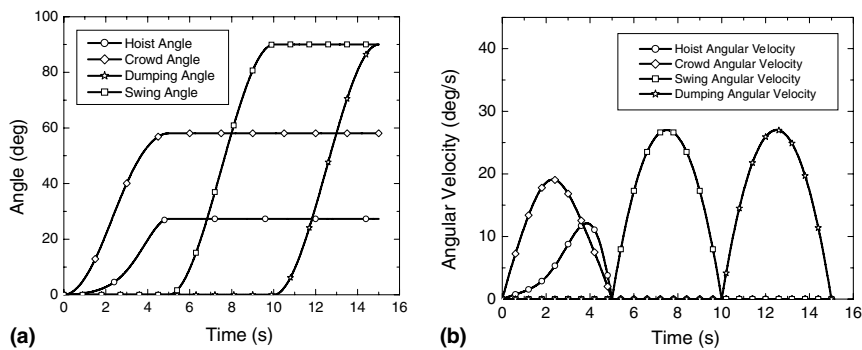


Fig. 6. Joint motion with time: (a) angles and (b) angular velocities [5].

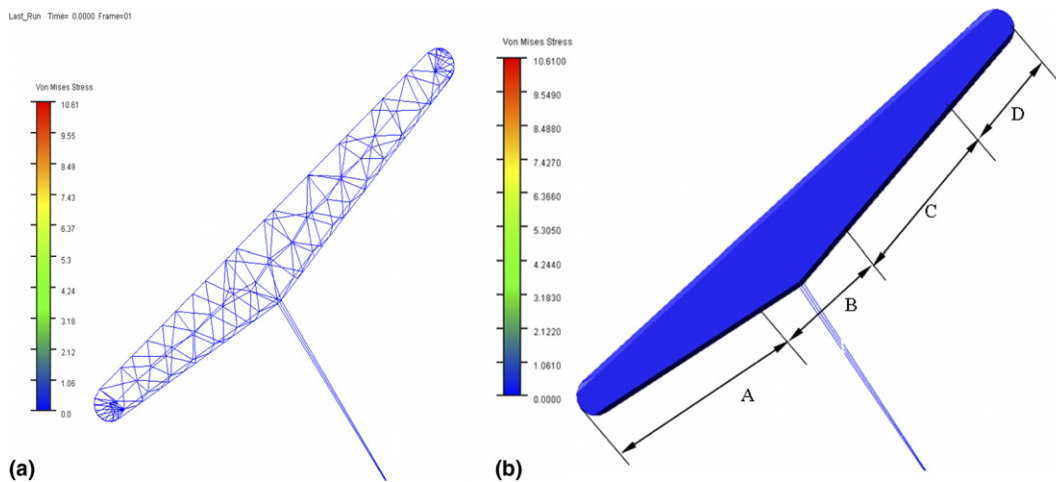


Fig. 7. Boom FE model: (a) isometric view and (b) shaded view.

operations (dipper-oil sands formation interactions) occur during crowding and hoisting operations. The loads applied on the boom are introduced by the hoist ropes and the crowd arm extensions and their interactions with the shovel's upper body and boom weight. The exact positions of the boom and the applied boom

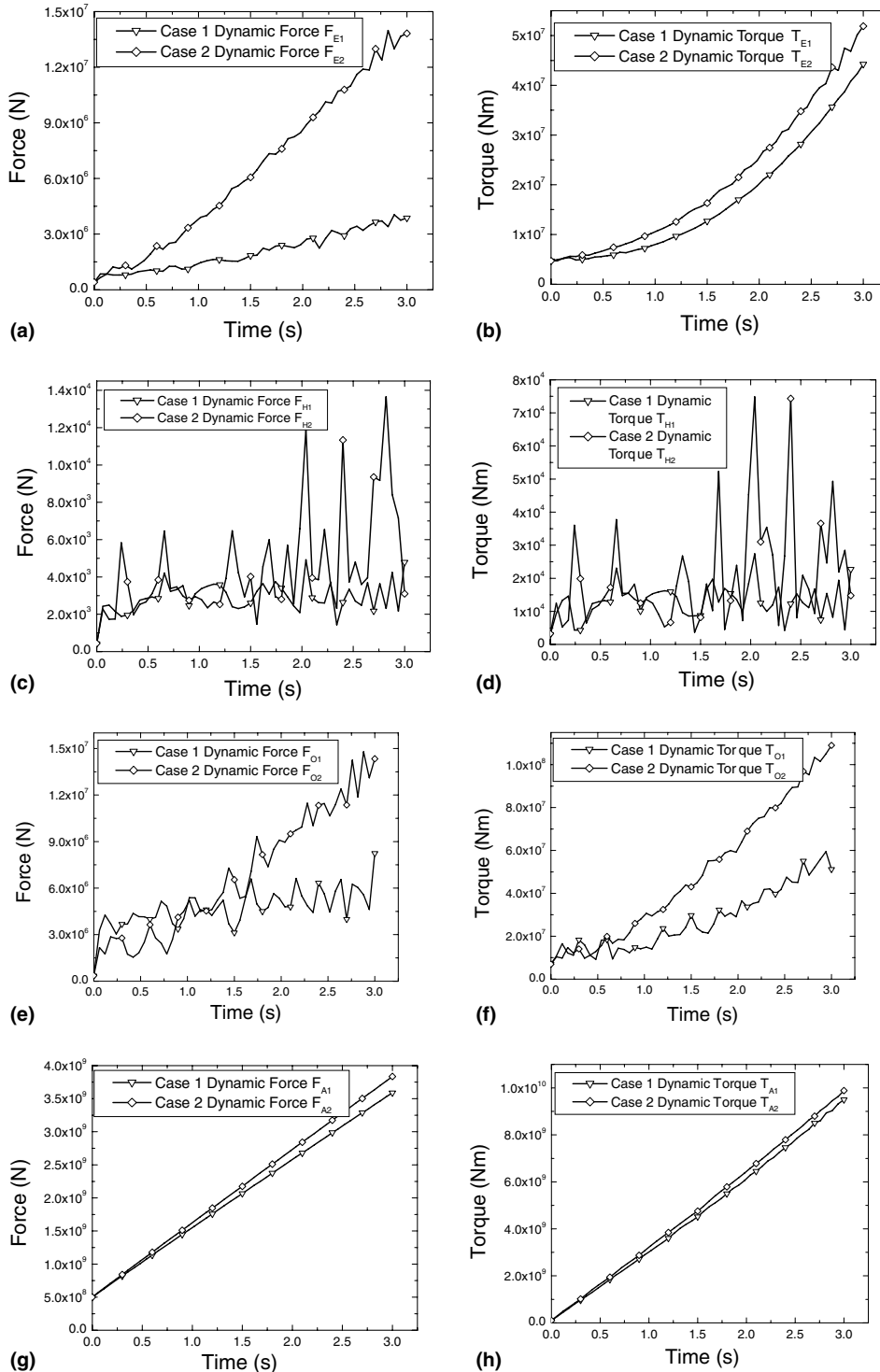


Fig. 8. Dynamic load applied to boom in Cases 1 and 2: (a) force F_E , (b) torque T_E , (c) force F_H , (d) torque T_H , (e) force F_O , (f) torque T_O , (g) force F_A , and (h) torque T_A .

forces are obtained from dynamic simulation in ADAMS. These forces are reflected in the dynamic FE model of the boom resulting from the oil sands material properties for the two cases considered in this study as suggested by Syncrude [14].

The simulation of the boom stress of 50 time steps are obtained for the crowding phase within 3 s using ADAMS. The cable shovel motion is visualized for an accurate prediction of the dynamic boom stress. Fig. 8 gives the dynamic loads applied to the boom at points E, H, O and A during digging time of 3 s using the oil sands materials for Cases 1 and 2 in Table 3. Fig. 8a, c, e and g illustrates the dynamic forces applied to the boom at points E, H, O and A, respectively, for these cases. In these figures, the dynamic forces at points E, H, O and A increase with times from 0 to 3 s. The minimum forces on all these points occur at zero time, while the maximum occur at 3 s. However, for point H in case 2, the forces oscillate with times from 0 to 3 s as illustrated in Fig. 8c, with a minimum force at zero and a maximum at 2.5 s. Fig. 8b, d, f and h indicates the dynamic torques applied to the boom at points E, H, O and A, respectively. These figures show that the dynamic torques at points E, H, O and A increase with time within the simulation period. The minimum torque occurs at zero time, while the maximum at 3 s. The torques at point H fluctuate significantly within this duration, with a minimum torque at zero and a maximum at 1.8 s as shown in Fig. 8d.

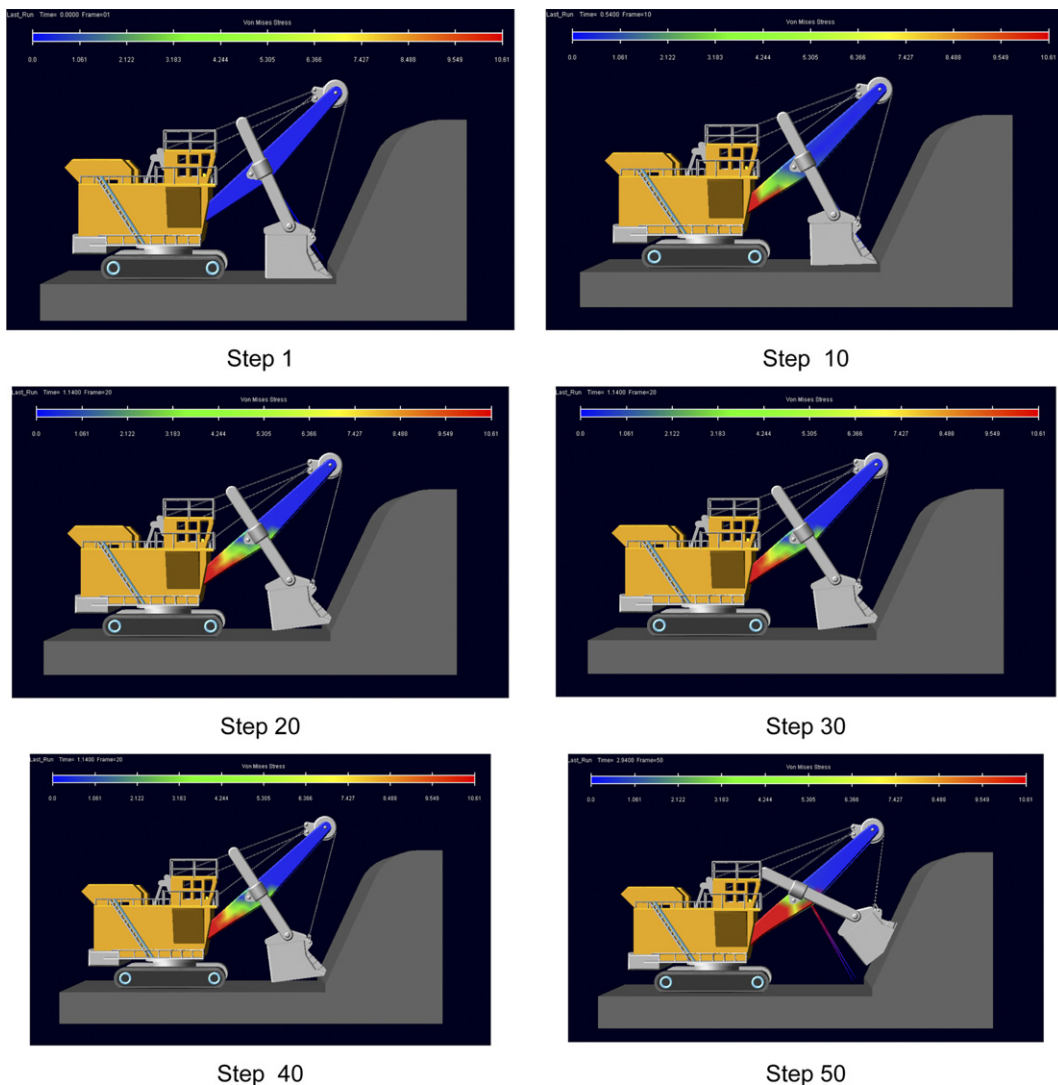


Fig. 9. Von Mises stress (GPa) contours on boom in Case 1.

The force and torque variations at point H are due to the variations in the spring loading of the hoist cables and the suspended load at the boom point sheave. The variations at point O result from the variations in the crowd arm loading due to its extension and digging operations. At point A, the boom is securely attached to the main shovel body, and thus, there are negligible variations in force and torque at this point even though the maximum occurs at this point. Fig. 9 shows the Von Mises stress contours of the boom zones A, B, C and D (Fig. 7b) for time steps 1–50 for Case 1. For any given time step, the maximum boom stress field occurs at Zone A and it decreases from zone A to zone B. The stress fields in zones C and D are smaller than in zones A and B. The stress field in zone B is spread over a complex geometrical shape, and it increases from time steps 1 to 50. Fig. 10 depicts the Von Mises stress distributions of the boom component from time steps 1 to 50 during 3 s for Case 2. For a given time step, the boom stress fields decrease from zone A to zone B with the maximum stress at zone A and minimum stresses in zones C and D are smaller than in zones A and B.

The stress distributions for both cases in Figs. 9 and 10 are similar at each time step. Figs. 9 and 10 also reveal the highly stressed FE nodes. The Von Mises stresses for the six highly stressed nodes are labeled in Fig. 11 around zone A. The corresponding Von Mises stresses acting at these nodes are shown in Table 4. The maximum stress value of 9.9 GPa occurs at node 178 for Case 1 and a maximum value of 10.61 GPa

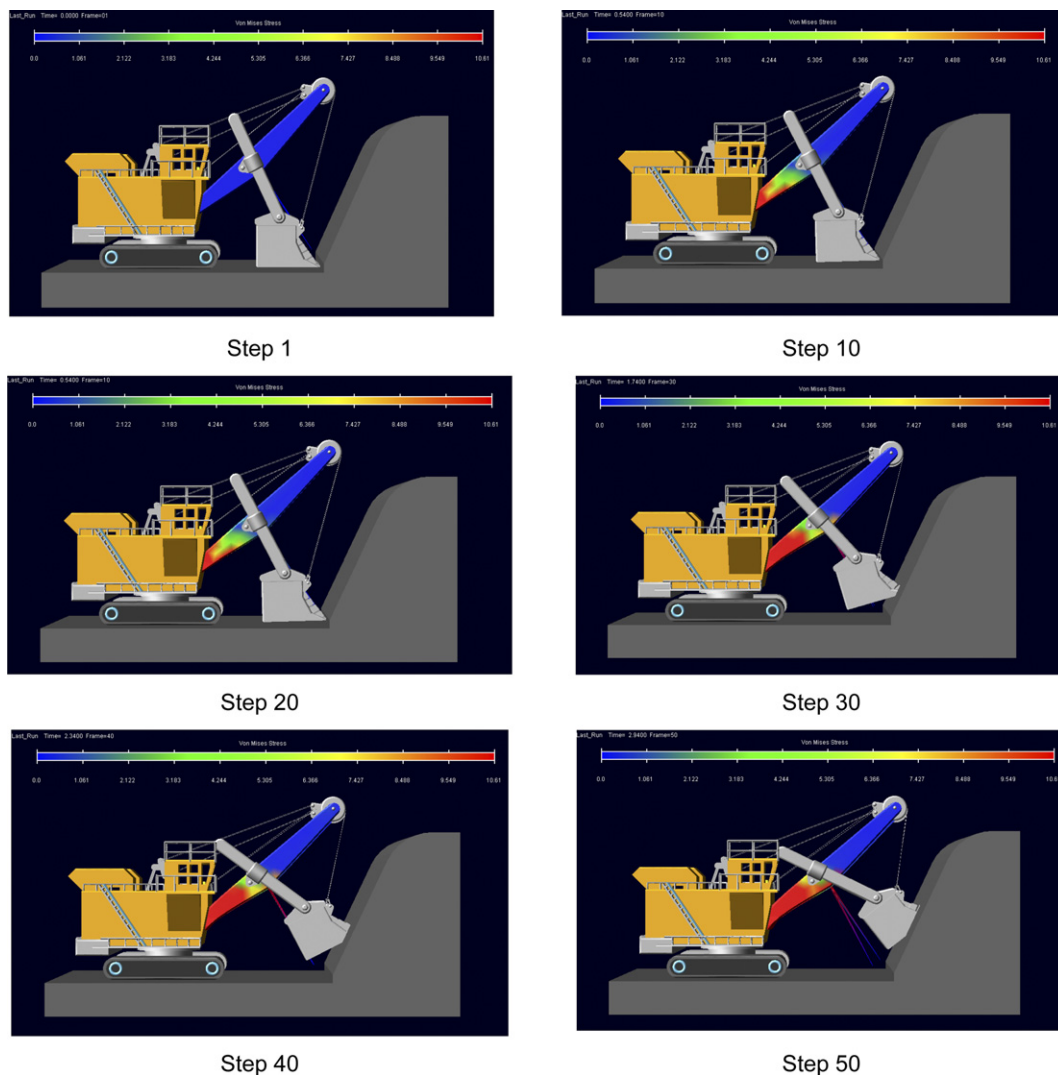


Fig. 10. Von Mises stress (GPa) contours on boom in Case 2.

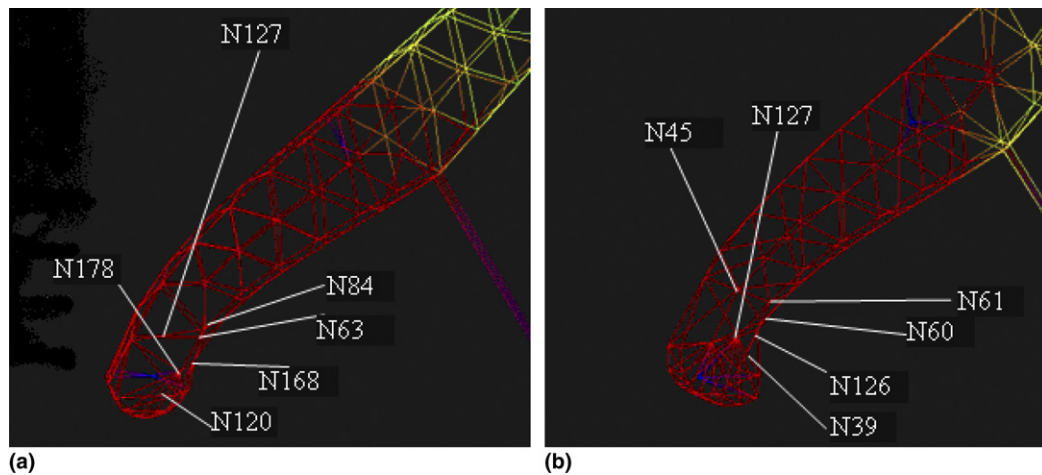


Fig. 11. Location of the nodes with top 6 hot spots at the web near A area of the boom at step 50: (a) Case 1 and (b) Case 2 at scale factor 12.

Table 4
Von Mises hot spots for Cases 1 and 2 at time step 50

Highly stressed zones	Case 1		Case 2		Case 2 – Case 1
	Stress (GPa)	Node	Stress (GPa)	Node	Stress (GPa)
1	9.9	178	10.61	127	0.71
2	9.71	168	9.94	61	0.23
3	9.61	120	9.91	126	0.30
4	9.55	63	9.77	45	0.22
5	9.46	127	9.62	60	0.16
6	9.42	84	9.55	39	0.13

at node 127 for Case 2. The Von Mises stresses at the six highly stressed nodes for Case 2 are greater than Case 1 by an average of 0.3 GPa during the digging and crowding operations. This is because the modulus of elasticity of oil sands material, in Case 2, is greater than in Case 1, resulting in greater applied boom loads as illustrated in Fig. 8. The critical area occurs at zone A, where potential durability problems may occur due to large concentrations of stress distributions. Therefore, excessive crowding during the digging operation will lead to reduced boom life.

6. Shovel optimum design and operating envelope

The drive toward machine health and longevity goals must recognize and mitigate large concentrations of stress fields [15]. High boom stress fields occur around the joint A, which may result in severe deformations and fatigue failure problems. This issue must be considered in boom geometry design for long-term structural integrity and real-time monitoring of stress field concentrations. The high levels of stress concentrations must be below a critical threshold. This threshold is a function of the Von Mises yield stress, equipment age and the applied design safety factor for the boom structural integrity in the long term. The stress loading analysis yields an optimum operating envelope for shovel excavation. This envelope contains the combination of crowding, hoisting and penetration speeds that result in an optimum shovel performance given an optimum material fragmentation. This optimum envelope minimizes the stress loading of the boom, which ultimately maximizes machine life with reduced maintenance costs. Consideration may also be given to pre-fragmentation of hard formations prior to excavation and loading. Real-time stress field monitoring could provide information on these hard formations. This study is also useful for determining the highly stressed nodes within the highly stressed zones for placing sensors for realtime monitoring of boom stresses.

7. Conclusions

The cable shovel and the dipper-oil sands formation interaction are modeled as a multi-body kinematics and dynamics systems using Newton-Lagrangian dynamic theory. FEM is used to provide solutions to the dynamic non-linear events for component durability. Component mode synthesis is used to produce flexible boom bodies with FE mesh to identify areas of high stress. Von Mises stresses are calculated through the sum of pseudo-static stresses and modal acceleration stresses. Durability analysis comes in the form of real-time stresses of the boom, which vary with respect to time step. A virtual cable shovel prototype simulator has been developed to simulate the stress field concentrations during the crowding and digging operations. The results show that the angular displacements are very smooth for each motion, increasing rapidly from minimum to maximum and then staying constant at the maximum. The joint angular velocities are also very smooth, increasing rapidly from minimum to maximum and decreasing with almost the same gradients. The respective maximum angular displacements (degrees) for the hoist, crowd, swing and dump motions are 30, 60, 90 and 90. The respective maximum angular velocities (degrees per second) for the hoist, crowd, swing and dump motions are 12, 20, 28 and 30. The results show that high stresses occur at regions around the joint between the upper body and the boom, resulting in large deformations. The maximum boom stress loading is 9.9 GPa for oil sands modulus of elasticity of 10 MPa and 10.61 GPa at 20 MPa modulus of elasticity. The results also show that the FE nodes of 178, 168, 120, 63, 127 and 84 for 10 MPa modulus of elasticity and 127, 61, 126, 45, 60 and 39 nodes for 20 MPa modulus of elasticity are the highly stressed nodes with high degree of boom deformation and fatigue failure.

Appendix A. Multi-body stress field recovery models

The motion of each shovel component in Fig. 1d is described in terms of the generalized coordinates, q_i , by Euler–Lagrangian formulation for constrained system by the following equation [11]:

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}_i} \right] - \frac{\partial L}{\partial q_i} = \tau_i; \quad i = 1, 2, \dots, n \quad (\text{A.1})$$

where L is Lagrangian $= T - V$, T is the kinetic energy of the system, V is the potential energy of the system, τ_i , generalized torques/forces applied to the system at joint i to drive pipeline i . The generalized coordinates are given by $\{q_i\}^T = \{x_i, y_i, \theta_i\}$, where the rotational components are the Euler angles in body 3-1-3 successive rotations. The reaction torques or forces in the cable shovel are given by the summation term in Eq. (A.1) along each of the generalized coordinates. The oil sands load-deformation model is created as a continuous system with a finite number of units connected by spherical joints. Fig. 2 shows a unit model with two DOFs, which have been idealized to a simple mass–spring–dashpot system. The mass is equal to the mass of an oil sand unit and the spring represents the elastic properties of the oil sands. The spring stiffness k represents the elastic modulus, E , of the oil sands. The dashpot represents the damping characteristic of the oil sands. The oil sand unit is subjected to a contact force when the dipper moves on it. Based on Newton's second law [12], the deformation for the unit can be formulated as

$$[M]\{\ddot{X}\} + [C]\{\dot{X}\} + [K]\{X\} = \{F\} \quad (\text{A.2})$$

$$[M] = [m \ J] \quad (\text{A.3})$$

$$[C] = \begin{bmatrix} 2c & -c(l_1 - l_2) \\ -c(l_1 - l_2) & c(l_1^2 - l_2^2) \end{bmatrix} \quad (\text{A.4})$$

$$[K] = \begin{bmatrix} 2k & -k(l_1 - l_2) \\ -k(l_1 - l_2) & k(l_1^2 - l_2^2) \end{bmatrix} \quad (\text{A.5})$$

$$\{F\} = \{f(t) \ M(t)\} \quad (\text{A.6})$$

where $[M]$ is the mass matrix; m is the mass of oil sand unit; J is the mass moment of inertia about the center of gravity; $[C]$ is the dashpot coefficient matrix; $[K]$ is the stiffness matrix; $\{X\}$ is the displacement vector, $\{X\} = \{x\theta\}$, x is the linear displacement of the center of gravity, and θ is the angular displacement of the unit;

$\{F\}$ is the external force vector; $f(t)$ is the external force applied to the unit; and $M(t)$ is the moment applied to the unit. Flexible body model of the boom is developed based on the rigid multi-body model by integrating a FE flexible body into the shovel model. Integrated environments consisting of mechanical system and finite element models are important to include flexible boom in a rigid system so that overall hybrid model can be built for the cable shovel component. Component mode synthesis is an effective method of evaluating stress, which allows reducing the total number of degrees of freedom (DOFs) of a typical FE component used for detailed stress analysis, while preserving its local deformations with high level of accuracy. Flexible structural component motion with N DOFs and defined boundaries are described by a combination of P normal modes and S constraint modes. The structural component of the cable shovel can be modeled in dynamic equilibrium equation

$$M\ddot{X} + KX = F \quad (\text{A.7})$$

where M and K are the mass and stiffness matrices for the flexible component, X and F are, respectively, the displacement vector at nodal coordinate and the load vector. The system DOFs are partitioned between internal and boundary DOFs, so the flexible body motion is given by

$$\begin{bmatrix} m^{BB} & 0 \\ 0 & m^{II} \end{bmatrix} \begin{Bmatrix} \ddot{x}^B \\ \ddot{x}^I \end{Bmatrix} + \begin{bmatrix} k^{BB} & k^{BI} \\ k^{IB} & k^{II} \end{bmatrix} \begin{Bmatrix} x^B \\ x^I \end{Bmatrix} = \begin{Bmatrix} f^B \\ f^I \end{Bmatrix} \quad (\text{A.8})$$

where I is the internal DOFs (equal to $N-S$) and B is the boundary ones (equal to S). From a static equilibrium analysis, assuming that interior forces are set to zero, Eq. (A.8) becomes Eq. (A.9). The constraint modes matrix, $[\phi^C]$, is given by Eq. (A.10). From an eigen-value analysis, the orthogonalized Craig–Bampton modes $\{\phi^I\}$ is obtained and the normal modes matrix $[\phi^N]$ is given by Eq. (A.11). From Eq. (A.11), a subset of the N normal modes is considered, and the physical coordinates are calculated as a linear combination of the mode shapes

$$\begin{bmatrix} k^{BB} & k^{BI} \\ k^{IB} & k^{II} \end{bmatrix} \begin{Bmatrix} x^B \\ x^I \end{Bmatrix} = \begin{Bmatrix} f^B \\ f^I \end{Bmatrix} \quad (\text{A.9})$$

$$[\phi^C] = -[k^{II}]^{-1}[k^{IB}] \quad (\text{A.10})$$

$$[\phi^N] = [\{\phi^I\}_I, \dots, \{\phi^I\}_P] \quad (\text{A.11})$$

$$\{x\} = \begin{Bmatrix} x^B \\ x^I \end{Bmatrix} = \begin{bmatrix} [I] & [0] \\ [\phi^C] & [\phi^N] \end{bmatrix} \begin{Bmatrix} p^B \\ p^I \end{Bmatrix} = [\Phi]\{p\} \quad (\text{A.12})$$

Eq. (A.8) can be rewritten as Eq. (A.13). Assuming that the reduction of the full set of mode shapes of the flexible body to subset is correct and captures all the required effects, the stress distribution related to the body deformation can be calculated in a similar way to the one used for physical displacements. $[\Phi_\sigma]$ is the full ortho-normalized modal stress matrix, and $\{\phi_\sigma\}_I$ is the $P+S$ and $\{p\}$ is the vector of modal coordinates.

$$[\bar{M}]\{\ddot{p}\} + [\bar{K}]\{p\} = \begin{bmatrix} \bar{m}^{BB} & \bar{m}^{BN} \\ \bar{m}^{NB} & \bar{m}^{NN} \end{bmatrix} \begin{Bmatrix} \ddot{p}^B \\ \ddot{p}^I \end{Bmatrix} + \begin{bmatrix} \bar{k}^{BB} & 0 \\ 0 & \bar{k}^{NN} \end{bmatrix} \begin{Bmatrix} p^B \\ p^I \end{Bmatrix} = \begin{Bmatrix} \bar{f}^B \\ \bar{f}^I \end{Bmatrix} = \{\bar{f}\} \quad (\text{A.13})$$

$$\{\sigma\} = [\phi_\sigma]\{p\} \quad (\text{A.14})$$

$$[\Phi_\sigma] = [\{\phi_\sigma\}_I, \dots, \{\phi_\sigma\}_{P+S}] \quad (\text{A.15})$$

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